Yue Xue (薛岳)

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https://bradmoonuestc.github.io/

https://github.com/BradMoonUESTC

https://scholar.google.com/citations?user=6cdLQQQAAAAJ

SUMMARY

- Senior Security Engineer focused on blockchain security, software security, and LLM applications in security.
- An early LLM4Sec researcher and builder: co-author of foundational, highly-cited works (GPTScan, PropertyGPT), advancing a security methodology combining LLMs with program analysis/formal verification, and building Finite Monkey to industrialize AI smart contract auditing at scale.
- Owned internal anti-money laundering (AML) initiatives at OKX with end-to-end on-chain compliance experience, from fund monitoring and risk identification to alerting/interdiction and closed-loop handling.
- Experienced in smart contract auditing, static analysis, formal verification, and on/off-chain monitoring systems.
- Awards: ETHSecurity Badge holder; NDSS 2025 Distinguished Paper Award; ACM SIGSOFT Distinguished Paper Award (FSE 2024); Finite Monkey ranked #2 on Secure3, earning USD 68,570 in cumulative bounties.

EDUCATION

Guangzhou University (GZHU) 2020.09 – 2023.06

M.S. in Computer Technology (Academician Fang Binxing Class)

· Research: blockchain security, smart contract security, LLM for security

University of Electronic Science and Technology of China (UESTC) 2011.09 – 2015.06

B.E. in Electrical Engineering and Automation

· Focus: electrical engineering, automatic control, and core engineering coursework

EXPERIENCE

CertiK 2025.11 – Present

Head of AI Research

· **Product engineering:** Lead developer of AI Auditor, a multi-language AI vulnerability scanning SaaS product spanning code understanding, threat modeling, vulnerability discovery, validation, and reporting.

· **Core capabilities:** Build finding deduplication, validation, and self-evolving workflows; optimize model usage, context reuse, and parallel execution to reduce costs and improve reliability.

· **Evaluation:** Built evaluation and regression workflows. Against 644 medium-or-higher findings in historical ACC human audit reports, achieved 80.1% exact-match coverage and 88.2% coverage including partial matches (August 27, 2026).

· **Competition results:** In an October 2026 Sherlock contest, the product tied for #1 among AI entrants in valid findings (22), tied for #1 overall in unique findings (7), and ranked #1 overall in coverage efficiency.

OKX 2022.04 – 2022.10, 2024.05 – 2025.11

Senior Security Engineer

· **AML risk control:** Owned ATMS (Asset Transaction Monitoring System) development and operations for first-line external-deposit risk control, covering illicit-fund detection and interception across CeFi deposits and wallet transactions.

· **On-chain investigations:** Led high-risk address tracing, linked-address expansion, intelligence maintenance, and risk assessment; coordinated manual investigations, fund-flow analysis, and risk reporting.

· **Team leadership:** Led a four-person team building smart contract monitoring for state and transaction anomalies; contributed to contract audits and security tooling.

· **Platforms and permissions:** Designed and maintained the internal contract management platform and privileged-task framework, completing permission migrations with zero incidents.

MetaTrust 2022.10 – 2025.08

Security Supervisor / Consultant

- **Audit delivery:** Led smart contract audit delivery and team coordination, improving audit workflows and quality control.
- **Detection tools:** Developed 60+ Slither detectors and optimized nearly 100 rules, reducing the false-positive rate below 25%.
- **AI security research:** Contributed to GPTScan and independently built PropertyGPT, combining LLMs with program analysis and formal verification.
- **Research collaboration:** Collaborated with NTU research teams and co-authored publications.

C# Developer 2016.12 – 2019.01

- Built full-stack web features and maintained databases
- Developed and maintained enterprise systems

PROJECTS & TOOLS

LLM4Sec Audit Tools Ongoing

End-to-end audit toolchain

- Code Audit Hinter (IDE security plugin, 39 stars, 2025.03–Present): RAG-powered audit hints for VSCode and Cursor
- Finite Monkey Engine (360 stars, 2024.04–Present): AI vulnerability scanning product built on a task-driven LLM audit engine
- TxAnalyzer (GitHub, 2026.05–Present): agent skill and CLI for blockchain attack transaction root-cause analysis, with EVM/Solana artifact collection
- AI Audit Result Evaluator (private, 2025.05–Present): automated judging for contest findings vs. ground truth
- Code Semantic Search (private, 2024.12–Present): LLM-based codebase understanding and retrieval

Falcon 2022.12 – 2024.05

Enhanced Solidity static analysis framework

- Extended Slither with dozens of new detectors and improved precision
- Supports DeFi price manipulation, centralization risk, and complex logic bug detection

Finite Monkey Product Results Ongoing

AI vulnerability scanning and competitions

- Ranked #2 on Secure3, earning USD 68,570 in cumulative bounties and demonstrating AI scanning performance in real-world audits.

PUBLICATIONS

1. Y. Sun, D. Wu, **Yue Xue**, H. Liu, H. Wang, Z. Xu, X. Xie, Y. Liu. "GPTScan: Detecting Logic Vulnerabilities in Smart Contracts by Combining GPT with Program Analysis." Proc. ICSE, 2024.
2. Y. Sun, D. Wu, **Yue Xue**, H. Liu, W. Ma, L. Zhang, Y. Liu, Y. Li. "LLM4Vuln: A Unified Evaluation Framework for Decoupling and Enhancing LLMs' Vulnerability Reasoning." arXiv preprint arXiv:2401.16185, 2024.
3. W. Ma, D. Wu, Y. Sun, T. Wang, S. Liu, J. Zhang, **Yue Xue**, Y. Liu. "Combining Fine-Tuning and LLM-based Agents for Intuitive Smart Contract Auditing with Justifications." Proc. ICSE, 2025.
4. Y. Liu, **Yue Xue***, D. Wu, Y. Sun, Y. Li, M. Shi, Y. Liu. "PropertyGPT: LLM-driven Formal Verification of Smart Contracts through Retrieval-Augmented Property Generation." NDSS, 2025.
5. K. Li, **Yue Xue**, S. Chen, H. Liu, K. Sun, M. Hu, H. Wang, Y. Liu, Y. Chen. "Static Application Security Testing (SAST) Tools for Smart Contracts: How Far Are We?" Proc. ACM Softw. Eng. (FSE), 1, 1447–1470, 2024.
6. **Yue Xue**, J. Fu, S. Su, Z. A. Bhuiyan, J. Qiu, H. Lu, N. Hu, Z. Tian. "Preventing Price Manipulation Attack by Front-Running." Proc. AAIS, 309–322, 2022.
7. M. Xie, M. Hu, Z. Kong, C. Zhang, Y. Feng, H. Wang, **Yue Xue**, H. Zhang, Y. Liu, et al. "DeFort: Automatic Detection and Analysis of Price Manipulation Attacks in DeFi Applications." Proc. ISSTA, 2024.
8. **Yue Xue**, D. Fan, S. Su, J. Fu, N. Hu, W. Liu, Z. Tian. "A Review on the Security of the Ethereum-Based DeFi Ecosystem." CMES–Computer Modeling in Engineering & Sciences, 139(1), 2024.
9. S. Su, Y. Tan, **Yue Xue**, C. Wang, H. Lu, Z. Tian, C. Shan, X. Du. "Detecting Smart Contract Project Anomalies in Metaverse." Proc. IEEE MetaCom, 2023.
10. Z. Kong, C. Zhang, M. Xie, M. Hu, **Yue Xue**, Y. Liu, H. Wang, Y. Liu. "Smart Contract Fuzzing Towards Profitable Vulnerabilities." Proc. ACM Softw. Eng. (FSE), 2, 153–175, 2025.

11. S. Su, **Yue Xue**, L. Lin, C. Wang, H. Lu, J. Qiu, Y. Sun, Y. Liu, Z. Tian. "Smart Contract Firewall: Protecting the On-Chain Smart Contract Projects." Proc. IEEE GLOBECOM, 5122–5128, 2023.

12. Y. Sun, S. Ding, L. Tan, **Yue Xue**, B. Li, C. Huang, S. Xiong. "Beyond Stars and Commits: A Fair and Manipulation-Resistant Framework for Measuring Developer Contributions." Proc. ACM FSE Companion, 2025.

13. J. Gao, **Yue Xue**, X. Xie, J. Cao, S. Thant, E. Lee, B. Xu. "Understanding Codebase like a Professional! Human—AI Collaboration for Code Comprehension."ICPC '26.

14. **Yue Xue**. "When and How Context Rot Appears in Coding Agents: A White-Box Study of Agent Skills in Code Auditing." arXiv '26 (preprint).

15. **Yue Xue**. "Chiral Analysis of Smart Contracts: Detecting Vulnerabilities from Relational Inconsistencies Across Business Paths."arXiv '26 (preprint).

16. Z. Kong, W. Xia, C. Wang, **Yue Xue**, Y. Lu, P. Li, S. Li, Z. Cao, Y. Liu. "Knowdit: Agentic Smart Contract Vulnerability Detection with Auditing Knowledge Summarization."arXiv '26 (preprint).

17. R. Liang, J. Chen, X. Li, H. Gu, Y. Feng, **Yue Xue**, C. Wu, Y. Liu. "EvoPoC: Automated Exploit Synthesis for DeFi Smart Contracts via Hierarchical Knowledge Graphs."arXiv '26 (preprint).

* Co-first author.

ACADEMIC SERVICE

- Reviewer: IEEE Transactions on Software Engineering (TSE); ACM Transactions on Software Engineering and Methodology (TOSEM); ACM Transactions on Privacy and Security (TOPS)
- Reviewer: 1st Workshop on Interpreting Agent Behavior (IAB), NeurIPS 2026.

INDUSTRY WRITING

- Write on AI auditing, agent methodology, and on-chain attacks. The methodology series has been republished by DeFiHackLabs, translated by LearnBlockchain, and included in a yBorg preparation reading list.
- **AI Auditing Methodology: Everything You Need to Know (Part I)**: English on X (51,212 views); Chinese translation (737 reads).
- **AI Auditing Methodology: Everything You Need to Know (Part II)**: English on X (16,812 views); Chinese translation (566 reads).
- **AI Auditing Methodology: Agent Self-Evolution – Drift, Reverse Evolution and Solutions** (June 2026); X version (19,936 views).
- **CertiK AIAuditor: Some Thoughts on AI Audit Methodology** (August 2026).
- **Why Your Skill Keeps Ignoring Instructions** (August 2026).

View and read counts retrieved October 14, 2026.

MEDIA INTERVIEW

- Interviewed expert: International Financial News, August 18, 2026, Repeated Boundary Violations: Is AI Getting Out of Control?, discussing AI agent safety boundaries and safeguards.

AWARDS

- NDSS 2025 Distinguished Paper Award: PropertyGPT
- ACM SIGSOFT Distinguished Paper Award (FSE 2024): SAST Tools for Smart Contracts
- GLOBECOM 2023 Best Paper Award: Smart Contract Firewall
- ETHSecurity Badge holder
- Finite Monkey ranked #2 on Secure3, earning USD 68,570 in cumulative bounties